Introduction to MS Project

Labs – House Project

You are in charge of the delivery of certain Work Packages (WPs) in the construction of a two-storey house (ground floor and first floor). You will deliver the Substructure, the Superstructure and some of the Interior finishings.

1 Planning Stage (Lab 1)

Table 2 summarizes the information gathered during *planning*: the WBS, the precedence relationships, the estimated durations for each activity, the labour resources required for achieving those durations, and the non-labour resource costs to deliver each activity.

The project's start date is 03/01/2022.

Labour cost information is provided separately in Table 1.

The current company resource assignment schedule shows that 3 Superstructure Labourers (SL) and 3 Finishings Labourers (FL) maximum are available daily for the entire duration of the project.

Table 1: Labour cost information.

Resource Name	Resource Acronym	Cost / standard hour	Cost / overtime hour	Cost / use	Accruing type
Superstructure Labourer	SL	£30.00	£45.00	£0.00	Prorated
Finishings Labourer	FL	£20.00	£30.00	£0.00	Prorated

Table 2: Project activity information for the House project (predecessors, durations, labour resources and non-labour costs) at the Planning stage.

WBS	ID	Description	Predecessors	Duration (days)	Labour resources	Non-labour costs
A		***Substructure***				
A.1	A	Foundations	-	6	3 SL[1]	£7,000
A.2	B	Basement Slab	A	4	2 SL	£5,500
B.		***Ground Floor***				
B.1		***Superstructure***				
B.1.1	C	Structural walls Ground Floor	B	8	3 SL	£10,000
B.1.2	F	Windows + Doors Ground Floor	H	2	2 SL	£3,500
B.2		***Interior***				
B.2.1	K	Stairs	H	1	3 FL[2]	£1,500
B.2.2	I	Wall partitions Ground Floor	H	4	2 FL	£4,000
B.2.3	L	Tiling Ground Floor	N,P	3	2 FL	£3,000
B.2.4	N	Wall painting Ground Floor	I	3	2 FL	£500
B.2.5	P	Ceiling painting Ground Floor	I	4	2 FL	£300
C		***1st Floor***				
C.1		***Superstructure***				
C.1.1	D	Slab 1st Floor	C	8	3 SL	£8,000
C.1.2	E	Structural walls 1st Floor	D	9	3 SL	£10,000
C.1.3	G	Windows + Doors 1st Floor	K	1	2 SL	£3,500
C.2		***Interior***				
C.2.1	J	Wall partitions 1st Floor	K	4	2 FL	£4,000
C.2.2	M	Carpeting 1st Floor	O,Q	3	2 FL	£3,000
C.2.3	O	Wallpaper 1st Floor	J	2	2 FL	£400
C.2.4	Q	Ceiling painting 1st Floor	J	4	2 FL	£300
D	H	Roof	E	7	3 SL	£25,000
E	R	Cleanup and Clear Site	F,G,L,M	1	2 FL	£0

[1] Superstructure Labourer
[2] Finishings Labourer

2 Project Monitoring Stage (Lab 2)

Table 3 summarizes the information gathered during the monitoring stage, and more specifically on Friday 4th March 2022. The table contains work progress percentages as well as any delayed task start.

Actual labour costs have also been reported different than planned, and are presented in Table 4.

Table 3: Project progress information as of Friday 4th March 2022

WBS	ID	Description	Work Progress	Actual Start
-		***House***		
A		***Substructure***		
A.1	A	Foundations	100%	-
A.2	B	Basement Slab	100%	-
B.		***Ground Floor***		
B.1		***Superstructure***		
B.1.1	C	Structural walls Ground Floor	100%	-
B.1.2	F	Windows + Doors Ground Floor	100%	-
B.2		***Interior***		
B.2.1	K	Stairs	65%	-
B.2.2	I	Wall partitions Ground Floor	25%	-
B.2.3	L	Tiling Ground Floor	0%	-
B.2.4	N	Wall painting Ground Floor	0%	-
B.2.5	P	Ceiling painting Ground Floor	0%	-
C		***1st Floor***		
C.1		***Superstructure***		
C.1.1	D	Slab 1st Floor	100%	-
C.1.2	E	Structural walls 1st Floor	100%	-
C.1.3	G	Windows + Doors 1st Floor	0%	-
C.2		***Interior***		
C.2.1	J	Wall partitions 1st Floor	0%	-
C.2.2	M	Carpeting 1st Floor	0%	-
C.2.3	O	Wallpaper 1st Floor	0%	-
C.2.4	Q	Ceiling painting 1st Floor	0%	-

D	H	Roof	100%	-
E	R	Cleanup and Clear Site	0%	-

Table 4: Actual labour cost information.

Resource Name	Resource Acronym	Cost / standard hour	Cost / overtime hour	Cost / use	Accruing type
Superstructure Labourer	SL	£28.00	£45.00	£0.00	Prorated
Finishings Labourer	FL	£20.00	£30.00	£0.00	Prorated